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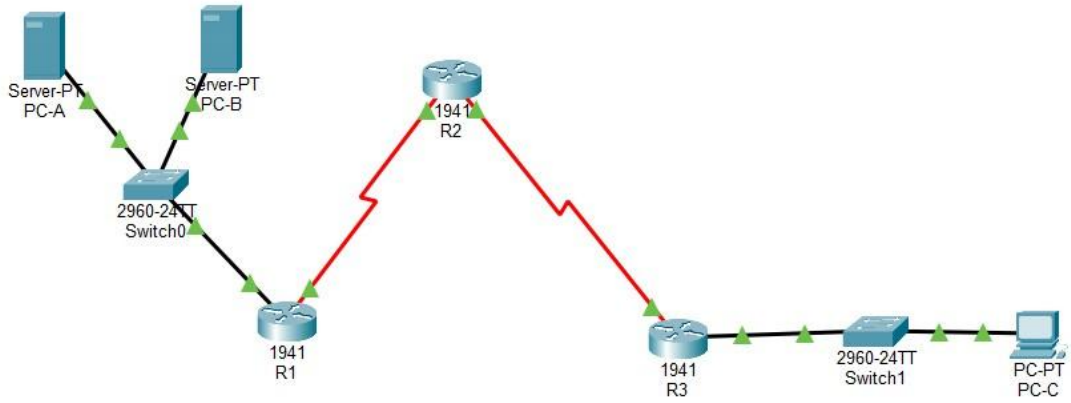
Practical 1 Topology:



Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

Part 1: Configure OSPF MD5 Authentication
Step 1: Configure OSPF MD5 authentication for all routers in area 0

- R1:
```
Router(config)#router ospf 1
Router(config-router)#area 0 authentication message-digest
Router(config-router)#exit
```
- R2:
```
Router(config)#router ospf 1
Router(config-router)#area 0 authentication message-digest
Router(config-router)#exit
```
- R3:

```
Router(config)#router ospf 1
Router(config-router)#area 0 authentication message-digest
Router(config-router)#exit
```

Step 2: Configure the MD5 key for all the routers in area 0

- R1:

```
Router(config)#interface se0/1/0
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#exit
```

- R2:

```
Router(config)#interface se0/1/0
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#exit
Router(config)#interface se0/1/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#exit
```

- R3:

```
Router(config)#interface se0/1/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#exit
```

Step 3: Verify the MD5 authentication

```
Router#show ip ospf interface
```

output:

```
    Adjacent with neighbor 10.2.2.2
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
Router#
```

Part 2 Configure NTP

Step 1: Enable NTP authentication on PC-A

a) On PC-A, click NTP under the Services tab to verify NTP service is enabled. Step 2:

Configure R1,R2, and R3 as NTP clients

- R1:

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp server 192.168.1.5
Router(config)#exit
```

- R2:

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp server 192.168.1.5
Router(config)#exit
```

- R3:

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp server 192.168.1.5
Router(config)#exit
```

Step 3: Configure routers to update hardware clock

- R1:

```
Router(config)#ntp update-calendar
Router(config)#
```

- R2:

```
Router(config)#ntp update-calendar
Router(config)#
```

- R3:

```
Router(config)#ntp update-calendar
Router(config)#
```

Step 4 : Configure NTP authentication on the routers •

R1:

```
Router(config)#ntp authenticate
Router(config)#ntp trusted-key 1
Router(config)#ntp authentication-key 1 md5 NTPpa55
```

- R2:

```
Router(config)#ntp authenticate
Router(config)#ntp trusted-key 1
Router(config)#ntp authentication-key 1 md5 NTPpa55
```

- R3:

```
Router(config)#ntp authenticate
Router(config)#ntp trusted-key 1
Router(config)#ntp authentication-key 1 md5 NTPpa55
```

Step 5 : Configure routers to timestamp log messages

- R1:

```
Router(config)#service timestamps log datetime msec
```

- R2:

```
Router(config)#service timestamps log datetime msec
```

- R3:

```
Router(config)#service timestamps log datetime msec
```

Step 6 : verify output

```
Router#show ntp associations

address         ref clock       st   when     poll    reach  delay          offset          disp
*~192.168.1.5   127.127.1.1     1    47       64      377    0.00           0.00            0.48
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
Router#
```

Part 3: Configure Routers to Log Messages to the Syslog Server

Step 1 : Configure the routers to remote host Syslog server that will receive the logging message

- R1:

```
Router(config)#logging host 192.168.1.6
Router(config)#
```

- R2:

```
Router(config)#logging host 192.168.1.6
Router(config)#
```

- R3:

```
Router(config)#logging host 192.168.1.6
Router(config)#
```

Step 2 : Verify Logging configuration

```
Router#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
          0 flushes, 0 overruns, xml disabled, filtering disabled)
```

Step 3 : Examine logs of the Syslog Server

a) From the Services tab of the Syslog Server's dialogue box, select the Syslog services button. Observe the logging messages received from the routers.

Syslog

Service ◉ On ○ Off

	Time	HostName	Message
1	02.11.2026 09:13:47.022 PM	192.168.1.1	%SYS-5-CONFIG_I: Configured from console by console
2	02.11.2026 09:13:47.022 PM	192.168.1.1	: %SYS-6-LOGGINGHOST_STARTSTOP: ...

Part 4 : Configure R3 to support SSH Connections

Step 1 : Configure a domain name

- R3:

```
Router(config)#ip domain-name ccnasecurity.com
Router(config)#
```

Step 2 : Configure users for login to the SSH server on R3

- R3:

```
Router(config)#username vecna privilege 15 secret vecnapa55
```

Step 3: Configure the incoming vty lines on R3 •

R3:

```
Router(config)#line vty 0 4
Router(config-line)#login local
Router(config-line)#transport input ssh
```

Step 4: Erase existing key pairs on R3

- R3:

```
Router(config-line)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.

Router(config)#
```

Step 5 : Generate the RSA encryption key pai for R3.

- R3:

```
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#
```

Step 6 : Verify the SSH configuration

- R3:

```
R3#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R3#
```

Step 7 : Configure SSH timeouts and authentication parameters

- R3:

```
R3(config)#ip ssh time-out 90
R3(config)#ip ssh authentication-retries 2
R3(config)#ip ssh version 2
```

Step 8 : Attempt to connect to R3 via telnet from PC-C

Command Prompt of PC-C

```
C:\>telnet 192.168.3.1
Trying 192.168.3.1 ...Open

[Connection to 192.168.3.1 closed by foreign host]
C:\>
```

Step 9: Connect to R3 using SSH on PC-C(note after '–' it is l(as in lunch) the letter not 1) Command Prompt of PC-C

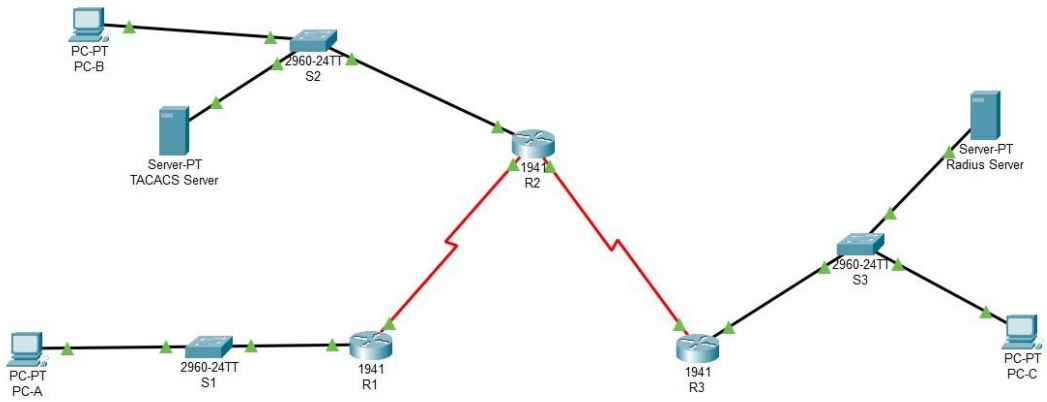
```
C:\>ssh -l vecna 192.168.3.1

Password:



R3#
```

Step 10 : Connect to R3 using SSH on R2

- R2:

```
Router>enable
Router#ssh -v 2 -l vecna 10.2.2.1

Password:



R3#
```

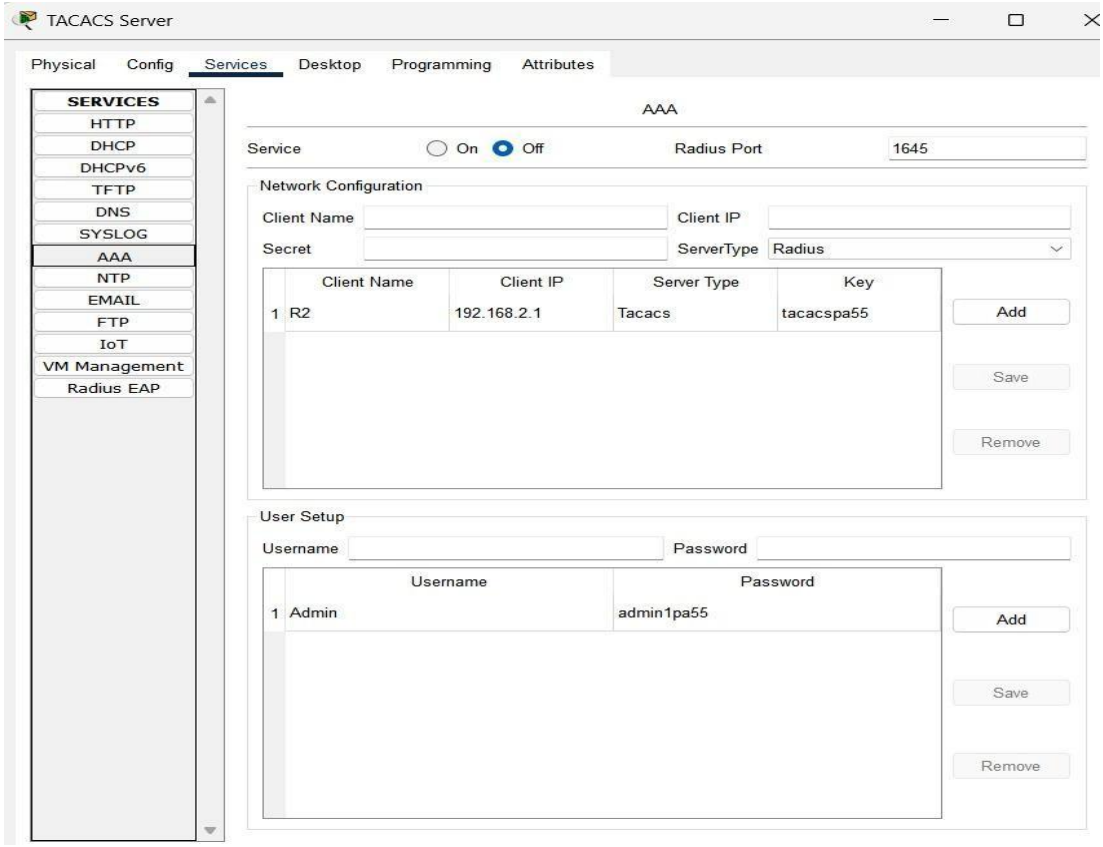
Practical 2 Topology:

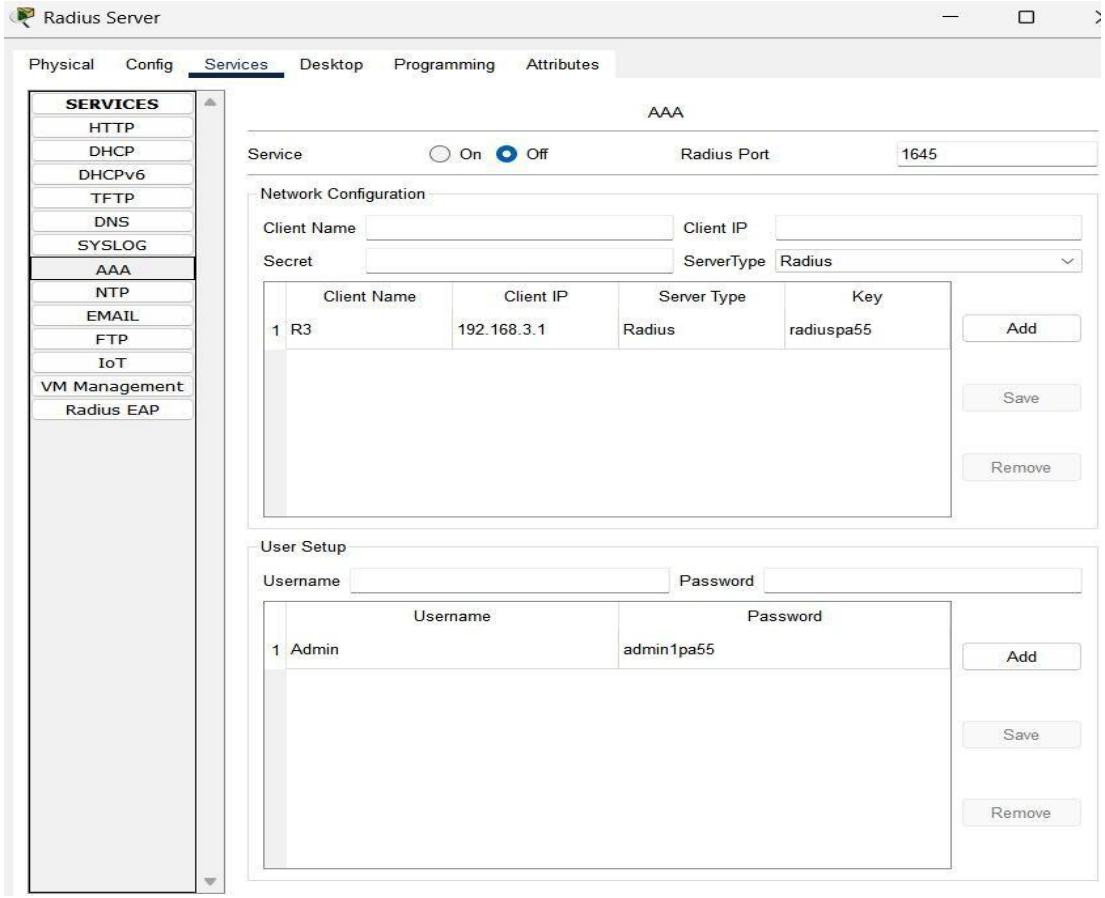


Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Server:





CLI commands:

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#
R1(config)#username Admin1 secret admin1pa55
R1(config)#
R1(config)#aaa new-model
R1(config)#aaa authentication login default local
R1(config)#line console 0
R1(config-line)#login authentication default
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#ex
```

```
User Access Verification

Username: Admin1
Password:
R1>
```

```
User Access Verification

Username: Admin1
Password:
R1>en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip domain-name ccnasecurity.co
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.co
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#aaa authentication login SSH-LOGIN local
*Mar 1 0:39:4.172: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login authentication SSH-LOGIN
R1(config-line)#transport input ssh
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

PC Outputs:

PC-A

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh 1 Admin1 192.168.1.1
Invalid Command.

C:\>ssh -l Admin1 192.168.1.1

Password:
R1>
```

CLI commands:

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
R2>
R2>en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#username Admin1 secret admin1pa55
R2(config)#tacacs-server host 192.168.2.2
R2(config)#tacacs-server key tacacspa55
R2(config)#aaa new-model
R2(config)#aaa authentication login default group tacacs+ local
R2(config)#line console 0
R2(config-line)#login authentication default
R2(config-line)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#exit
```

```
User Access Verification

Username: Admin
Password:
R2>
```

R3

Physical Config CLI Attributes

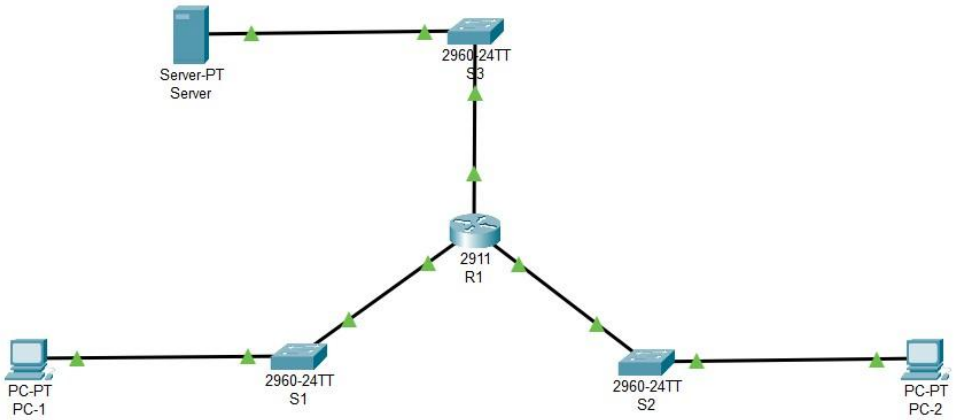
IOS Command Line Interface

```
Press RETURN to get started!


R3>en
R3#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#username Admin1 secret adminlpa55
R3(config)#radius-server host 192.168.3.2
R3(config)#radius-server key radiuspa55
R3(config)#aaa new-model
R3(config)#aaa authentication login default group radius local
R3(config)#line console 0
R3(config-line)#
R3(config-line)#login authentication default
R3(config-line)#
R3(config-line)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#exit
```

```
Username: Admin
Password:
R3>
```

Practical 3A

Topology:



Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.22.34.65	255.255.255.224	N/A
	G0/1	172.22.34.97	255.255.255.240	N/A
	G0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

CLI commands:

```
R1
Physical   Config   CLI   Attributes
IOS Command Line Interface
Router(config)#hostname R1
R1(config)#
%SYS-5-CONFIG_I: Configured from console by console

R1(config)#access-list ?
  <1-99>     IP standard access list
  <100-199>  IP extended access list
R1(config)#access-list
% Incomplete command.
R1(config)#access-list 100 ?
  deny    Specify packets to reject
  permit  Specify packets to forward
  remark  Access list entry comment
R1(config)#access-list 100 permit ?
  ahp    Authentication Header Protocol
  eigrp  Cisco's EIGRP routing protocol
  esp    Encapsulation Security Payload
  gre    Cisco's GRE tunneling
  icmp   Internet Control Message Protocol
  ip     Any Internet Protocol
  ospf   OSPF routing protocol
  tcp    Transmission Control Protocol
  udp    User Datagram Protocol
R1(config)#access-list 100 permit tcp ?
  A.B.C.D  Source address
  any      Any source host
  host     A single source host
R1(config)#access-list 100 permit tcp 172.22.34.64 ?
  A.B.C.D  Source wildcard bits
```

```
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 ?
  A.B.C.D  Destination address
  any      Any destination host
  eq       Match only packets on a given port number
  gt       Match only packets with a greater port number
  host     A single destination host
  lt       Match only packets with a lower port number
  neq      Match only packets not on a given port number
  range    Match only packets in the range of port numbers
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 ?
  dscp         Match packets with given dscp value
  eq           Match only packets on a given port number
  established  established
  gt           Match only packets with a greater port number
  lt           Match only packets with a lower port number
  neq          Match only packets not on a given port number
  precedence   Match packets with given precedence value
  range        Match only packets in the range of port numbers
  <cr>
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ?
  <0-65535>  Port number
```

Copy Paste

Top

```
  <cr>
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ?
  <0-65535>  Port number
  ftp        File Transfer Protocol (21)
  pop3       Post Office Protocol v3 (110)
  smtp       Simple Mail Transport Protocol (25)
  telnet     Telnet (23)
  www        World Wide Web (HTTP, 80)
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
R1(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip access-group 100
% Incomplete command.
R1(config-if)#ip access-group 100 in
R1(config-if)#
```

PC Outputs:

```
C:\>ftp 172.22.34.62
Trying to connect...172.22.34.62

%Error opening ftp://172.22.34.62/ (Timed out)
.



(Disconnecting from ftp server)
```

PC-1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 172.22.34.62
Trying to connect...172.22.34.62
Connected to 172.22.34.62
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>quit

221- Service closing control connection.
C:\>
```

CLI commands:

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1>en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list ?
  extended  Extended Access List
  standard  Standard Access List
R1(config)#ip access-list
% Incomplete command.
R1(config)#ip access-list ?
  extended  Extended Access List
  standard  Standard Access List
R1(config)#ip access-list ?
  extended  Extended Access List
  standard  Standard Access List
R1(config)#ip access-list ip access-list extended HTTP_ONLY
                          ^
% Invalid input detected at '^' marker.

R1(config)#ip access-list extended HTTP_ONLY
R1(config-ext-nacl)# permit tcp 172.22.34.96 ?
  A.B.C.D  Source wildcard bits
R1(config-ext-nacl)# permit tcp 172.22.34.96  permit tcp 172.22.34.96 0.0.0.15 ?
% Unrecognized command
R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15 ?
  A.B.C.D  Destination address
  any      Any destination host
  eq       Match only packets on a given port number
  gt       Match only packets with a greater port number
  host     A single destination host
  lt       Match only packets with a lower port number
  neq      Match only packets not on a given port number
  range    Match only packets in the range of port numbers
R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15
% Incomplete command.
R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
R1(config-ext-nacl)# permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
R1(config-ext-nacl)#ex
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip access-group HTTP_ONLY in
R1(config-if)#
```

PC Outputs:

PC-2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.22.34.62

Pinging 172.22.34.62 with 32 bytes of data:

Reply from 172.22.34.62: bytes=32 time=12ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127
Reply from 172.22.34.62: bytes=32 time=2ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.22.34.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>
```

PC-2

Physical Config Desktop Programming Attributes

Web Browser

< > URL http://172.22.34.62

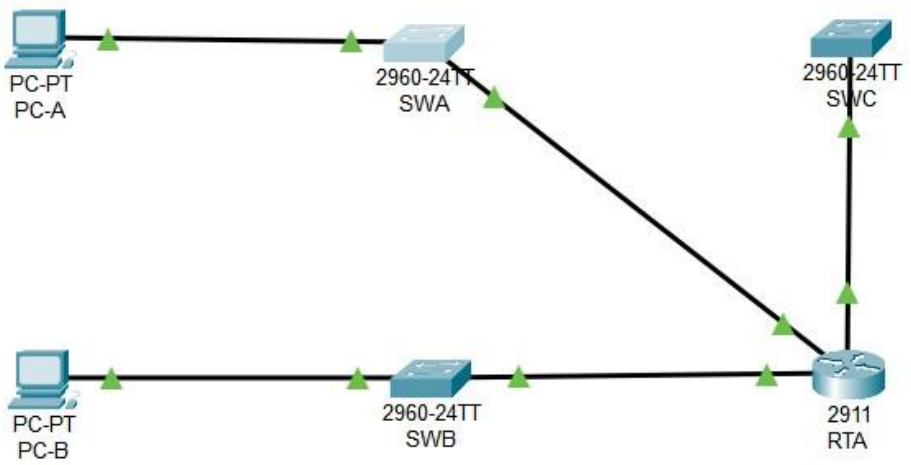
Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
A small page
Copyrights
Image page
Image

Practical 3B

Topology:



Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
RTA	G0/0	10.101.117.49	255.255.255.248	N/A
	G0/1	10.101.117.33	255.255.255.240	N/A
	G0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN 1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN 1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN 1	10.101.117.2	255.255.255.224	10.101.117.1

CLI Commands:

```
SWA>en
SWA#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SWA(config)#int vlan 1
SWA(config-if)#ip address 10.101.117.50 255.255.255.248
SWA(config-if)#no shutdown

SWA(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SWA(config-if)#ip default-gateway 10.101.117.49
SWA(config)#enable secret enpa55
SWA(config)#line console 0
SWA(config-line)#password conpa55
SWA(config-line)#login
SWA(config-line)#
```

```
SWB>en
SWB#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SWB(config)#int vlan 1
SWB(config-if)#ip address 10.101.117.34 255.255.255.240
SWB(config-if)#no shutdown

SWB(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SWB(config-if)#ip default-gateway 10.101.117.33
SWB(config)#enable secret enpa55
SWB(config)#line console 0
SWB(config-line)#password conpa55
SWB(config-line)#login
SWB(config-line)#
```

SWA

Physical Config CLI Attributes

IOS Command Line Interface

```
User Access Verification

Password:

SWA>en
Password:
SWA#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SWA(config)#line console 0
SWA(config-line)#en
% Ambiguous command: "en"
SWA(config)#
SWA(config)#ip domain-name ccnasecurity.com
SWA(config)#username Admin secret Adminpa55
SWA(config)#lint vty 0 4
                ^
% Invalid input detected at '^' marker.

SWA(config)#line vty 0 4
SWA(config-line)#login local
SWA(config-line)#crypto key generate rsa
The name for the keys will be: SWA.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SWA(config)#
*Mar 1 1:25:18.239: %SSH-5-ENABLED: SSH 1.99 has been enabled
SWA#
%SYS-5-CONFIG_I: Configured from console by console

SWA#
```

SWB

Physical Config CLI Attributes

IOS Command Line Interface

```
SWB>en
Password:
SWB#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SWB(config)#ip domain-name ccnasecurity.com
SWB(config)#username Admin secret Adminpa55
SWB(config)#line vty 0 4
SWB(config-line)#login local
SWB(config-line)#crypto key generate rsa
The name for the keys will be: SWB.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SWB(config)#





SWB con0 is now available
```

SWC

Physical Config CLI Attributes

IOS Command Line Interface

```
SWC>en
Password:
SWC#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SWC(config)#ip domain-name ccnasecurity.com
SWC(config)#username Admin secret Adminpa55
SWC(config)#line vty 0 4
SWC(config-line)#login local
SWC(config-line)#crypto key generate rsa
The name for the keys will be: SWC.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SWC(config)#





SWC con0 is now available
```

```
RTA>en
RTA#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RTA(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22
RTA(config)#access-list 199permit icmp any any
                          ^
% Invalid input detected at '^' marker.

RTA(config)#access-list 199 permit icmp any any
RTA(config)#ip access-group 199 out
                        ^
% Invalid input detected at '^' marker.

RTA(config)#interface Enter configuration commands, one per line.  End with CNTL/Z.
RTA(config)#Enter configuration commands, one per line.  End with CNTL/Z.
RTA(config)#
RTA(config)#interface GigabitEthernet0/2
RTA(config-if)#ip access-group 199 out
RTA(config-if)#
```

PC Output:

PCA

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>ping 10.101.117.35

Pinging 10.101.117.35 with 32 bytes of data:

Reply from 10.101.117.35: bytes=32 time<1ms TTL=127
Reply from 10.101.117.35: bytes=32 time<1ms TTL=127
Reply from 10.101.117.35: bytes=32 time<1ms TTL=127
Reply from 10.101.117.35: bytes=32 time<1ms TTL=127

Ping statistics for 10.101.117.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l Admin 10.101.117.34

Password:



SWB>exit

[Connection to 10.101.117.34 closed by foreign host]
C:\>ssh -l Admin 10.101.117.2

% Connection timed out; remote host not responding
C:\>
```

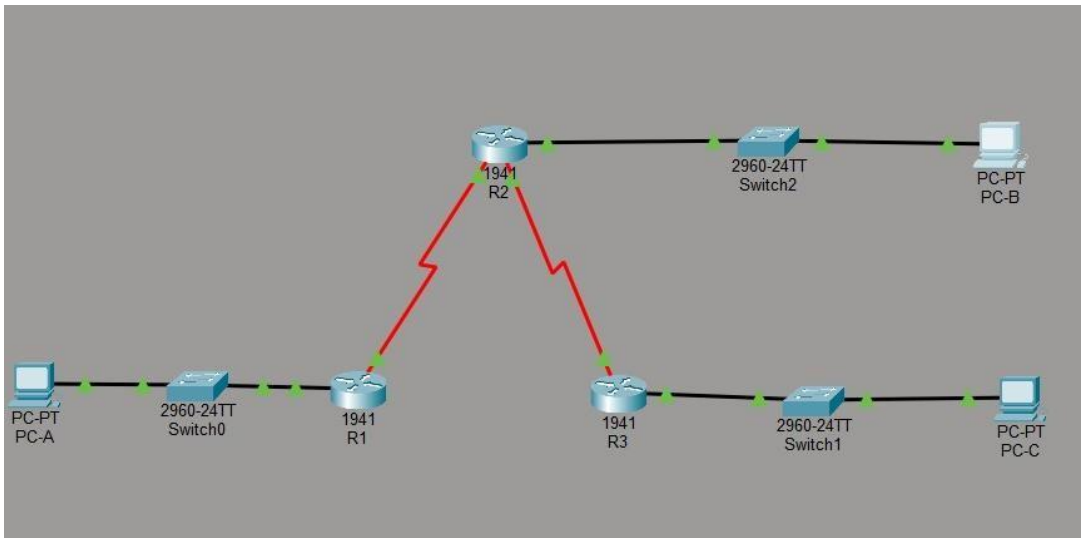
PCB

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>ping 10.101.117.2

Pinging 10.101.117.2 with 32 bytes of data:

Reply from 10.101.117.2: bytes=32 time<1ms TTL=254
Reply from 10.101.117.2: bytes=32 time<1ms TTL=254
Reply from 10.101.117.2: bytes=32 time<1ms TTL=254
Reply from 10.101.117.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.101.117.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l Admin 10.101.117.2

Password:



SWC>exit

[Connection to 10.101.117.2 closed by foreign host]
C:\>ssh -l Admin 10.101.117.50

Password:



SWA>exit

[Connection to 10.101.117.50 closed by foreign host]
C:\>
```

Practical 4 Topology:



Adressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/0	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

CLI Commands:

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
R1(config)#line console 0
R1(config-line)#password conpa55
R1(config-line)#login
R1(config-line)#exit
R1(config)#ip domain-name ccnasecurity.com
R1(config)#username admin secret adminpa55
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exit
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#
*Mar 1 0:0:41.675: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
*Mar 1 0:2:59.495: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

00:05:42: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on Serial0/1/0 from LOADING to FULL, Loading Done

R1(config-router)#exit
R1(config)#license boot module c1900 technology-package securityk9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

Activation  of the  software command line interface will be evidence of
your acceptance of this agreement.


ACCEPT? [yes/no]: y
% use 'write' command to make license boot config take effect on next boot

R1(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot level = securityk9 and
License = securityk9
exit
R1#
```

Copy | Paste

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
R1#
%SYS-5-CONFIG_I: Configured from console by console
reload
System configuration has been modified. Save? [yes/no]:y
Building configuration...
[OK]
Proceed with reload? [confirm]ySystem Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
################################################################################ [OK]
Smart Init is enabled
smart init is sizing iomem
                TYPE      MEMORY_REQ
           HWIC Slot 1      0x00200000     Onboard devices &
          buffer pools      0x01E8F000
----------------------------------------------
                TOTAL:      0x0268F000
Rounded IOMEM up to: 40Mb.
Using 6 percent iomem. [40Mb/512Mb]

              Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
           cisco Systems, Inc.
           170 West Tasman Drive
           San Jose, California 95134-1706

Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
```

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

00:00:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on Serial0/1/0 from LOADING to FULL, Loading Done


User Access Verification

Password:

R1>en
Password:
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key vpnpa55 address 10.2.2.2
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
```

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key vpnpa55 address 10.2.2.2
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#interface s0/1/0
R1(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
 show crypto ipsec sa

interface: Serial0/1/0
    Crypto map tag: VPN-MAP, local addr 10.1.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
     current outbound spi: 0x0(0)

     inbound esp sas:

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:

     outbound pcp sas:
```

R2

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
R2(config)#enable secret enpa55
R2(config)#line console 0
R2(config-line)#password conpa55
R2(config-line)#login
R2(config-line)#exit
R2(config)#ip domain-name ccnasecurity.com
R2(config)#username admin secret adminpa55
R2(config)#line vty 0 4
R2(config-line)#login local
R2(config-line)#exit
R2(config)#crypto key generate rsa
The name for the keys will be: R2.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R2(config)#router ospf 2
*Mar 1 0:5:57.694: %SSH-5-ENABLED: SSH 1.99 has been enabled
R2(config-router)#network 192.168.
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
2.0 0.0.0.255 area 0
R2(config-router)#network 10.2.2.0 0.0.0.3 area 0
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#
00:08:29: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from LOADING to FULL, Loading Done
```

R3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router(config)#hostname R3
R3(config)#
R3(config)#enable secret enpa55
R3(config)#line console 0
R3(config-line)#password conpa55
R3(config-line)#login
R3(config-line)#exit
R3(config)#ip domain-name ccnasecurity.com
R3(config)#username admin secret adminpa55
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#exit
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#
*Mar 1 0:0:35.909: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

R3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
*Mar 1 0:5:56.780: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#router ospf 3
R3(config-router)#network 192.168.3.0 0.0.0.255 area 0
R3(config-router)#network 10.2.2.0 0.0.0.3 area 0
R3(config-router)#
00:09:04: %OSPF-5-ADJCHG: Process 3, Nbr 192.168.2.1 on Serial0/1/1 from LOADING to FULL, Loading Done

R3(config-router)#exit
R3(config)#license boot module c1900 technology-package securityk9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

Activation  of the  software command line interface will be evidence of
your acceptance of this agreement.


ACCEPT? [yes/no]: y
% use 'write' command to make license boot config take effect on next boot

R3(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot level = securityk9 and
License = securityk9
exit
R3#
%SYS-5-CONFIG_I: Configured from console by console
reload
```

R3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
R3#
%SYS-5-CONFIG_I: Configured from console by console
reload
System configuration has been modified. Save? [yes/no]:y
Building configuration...
[OK]
Proceed with reload? [confirm]ySystem Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
################################################################### [OK]
Smart Init is enabled
smart init is sizing iomem
             TYPE     MEMORY_REQ
       HWIC Slot 1    0x00200000     Onboard devices &
      buffer pools    0x01E8F000
----------------------------------------------------
                TOTAL:      0x0268F000
Rounded IOMEM up to: 40Mb.
Using 6 percent iomem. [40Mb/512Mb]

              Restricted Rights Legend
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           170 West Tasman Drive
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Compiled Thurs 5-Jan-12 15:41 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

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```

R3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
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Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

00:00:20: %OSPF-5-ADJCHG: Process 3, Nbr 192.168.2.1 on Serial0/1/1 from LOADING to FULL, Loading Done


User Access Verification

Password:

R3>en
Password:
R3#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#group 5
R3(config-isakmp)#exit
R3(config)#crypto isakmp key vpnpa55 address 10.1.1.2
R3(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R3(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R3(config-crypto-map)#description VPN connection to R1
R3(config-crypto-map)#set peer 10.1.1.2
R3(config-crypto-map)#set transform-set VPN-SET
R3(config-crypto-map)#match address 110
R3(config-crypto-map)#exit
```
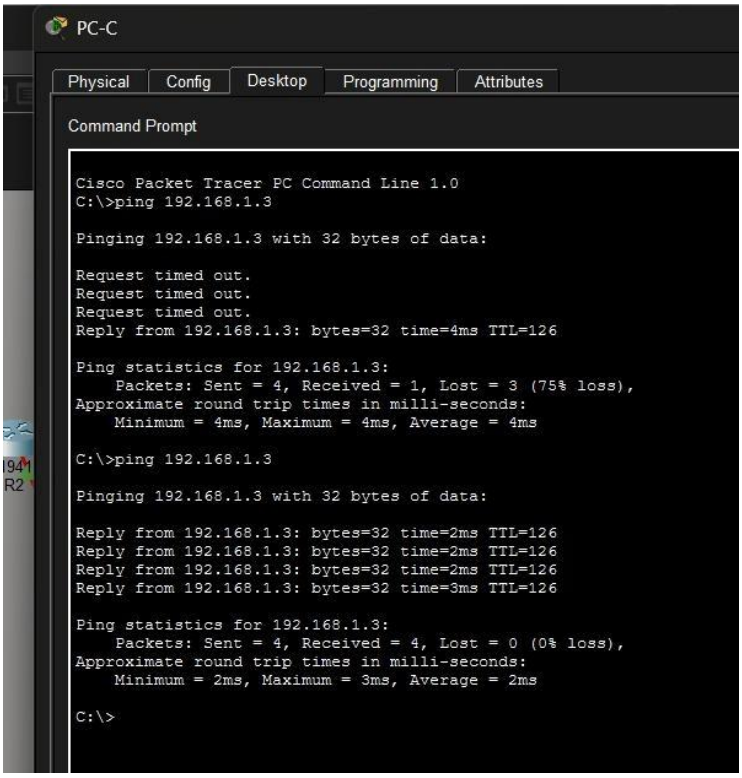

Copy | Paste

```
R3

Physical   Config   CLI   Attributes

IOS Command Line Interface

R3(config-crypto-map)#description VPN connection to R1
R3(config-crypto-map)#set peer 10.1.1.2
R3(config-crypto-map)#set transform-set VPN-SET
R3(config-crypto-map)#match address 110
R3(config-crypto-map)#exit
R3(config)#interface s0/1/1
R3(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R3(config-if)#exit
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3# show crypto ipsec sa

interface: Serial0/1/1
    Crypto map tag: VPN-MAP, local addr 10.2.2.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   current_peer 10.1.1.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.2.2.2, remote crypto endpt.:10.1.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/1
     current outbound spi: 0x0(0)

     inbound esp sas:

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:

     outbound pcp sas:

R3#
R3#
```

PC Outputs:

```
PC-A

Physical   Config   Desktop   Programming   Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=7ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 3ms

C:\>
```


PC-C
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.1.3: bytes=32 time=4ms TTL=126
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 4ms, Average = 4ms
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=3ms TTL=126
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\>